

Elliott Tai

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Education

Washington State University

Bachelor of Science in Computer Science, Minor in Mathematics
GPA

Pullman, WA

Expected May 2027
4.0

Certifications

AWS Certified Cloud Practitioner

Relevant Courses

Systems Programming C/C++, Programming Design and Development C/C++, Advanced, Data Structures, Object Oriented Software Principles, Software Engineering, Computer Architecture

Projects and Activities

Teaching AI to play video games | C#, Unity, ML-Agents, PyTorch

GitHub: <https://github.com/s-etai/Machine-learning-risky-road>

- Designed and programmed a Unity game to serve as a reinforcement learning test environment.
- Implemented agent logic, reward shaping, and training pipeline using Unity ML-Agents and PyTorch.
- Tuned the PPO algorithm to achieve AI performance $3\times$ higher than human baseline scores.

Spreadsheet app | C#, Windows Forms, .NET

GitHub: <https://github.com/s-etai/Spread-sheet-app>

- Developed a fully functional spreadsheet application demonstrating object-oriented design and software architecture principles.
- Implemented cell editing, formulas, expression evaluation, and cell references.
- Added **Undo/Redo** functionality using the **Command pattern** and **event-driven updates** via the **Observer pattern** (`INotifyPropertyChanged`).
- Designed a **layered architecture** with high cohesion and low coupling between UI, logic, and data components.

CPU scheduling simulator | C, Linux

- Implemented a CPU scheduling simulator in C supporting multiple algorithms including FCFS, SJF, and Round Robin.
- Parsed process data from input files and accurately calculated waiting time, turnaround time, and CPU utilization for each scheduling method.

- Designed modular functions and data structures to efficiently manage processes and scheduling queues.

Atari Classic Centipede recreation | C++, SFML

GitHub: <https://github.com/calebh13/PA9>

- Collaborated with a team of four to recreate the classic Atari game **Centipede** using **C++** and the **SFML graphics library**.
- We implemented core gameplay systems including player movement, projectile firing, enemy segmentation, collision detection, and score tracking.
- Strengthened **teamwork**, **problem-solving**, and **software architecture** skills through iterative development and testing

Leetcode club | Committee Chair

- Organized weekly technical sessions and collaborated with peers to solve algorithmic problems.

Association for Computing Machinery (ACM) | Active Member

- Participated in coding workshops and attended guest lectures.

Technical Skills

Languages: C, C++, C#, Python

Developer Tools: Git, Teamwork, Communication